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ABOUT ME

Senior Full-Stack Engineer with 7.5 years of experience delivering high-impact solutions across diverse industries, from EdTech to AI-driven advanced manufacturing. Highly adaptable and quick to master complex new domains, consistently driving business value through scalable architectures and robust code. Proficient in modern frontend development (React, JavaScript, TypeScript, Three.js), backend engineering (Python, Django, Ruby, PostgreSQL) and infrastructure (Docker, Kubernetes, GCP, AWS, Terraform). I thrive in fast-paced environments, taking full ownership of end-to-end features to solve real-world problems - from system design to cloud deployment and ongoing maintenance.

SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, Go, Kotlin, Ruby, SQL
- **Frontend, 3D Web:** React, React Three Fiber (R3F) and Drei, Three.js, Tailwind CSS, styled-components, HTML5, CSS3
- **Backend, APIs, Data:** Django, drf-spectacular (Swagger/OpenAPI), FastAPI, Flask, Gunicorn, PostgreSQL, Ruby on Rails
- **3D Processing, Analytics:** Dagster, CadQuery, OpenCascade, Trimesh, Shapely, NumPy, SciPy
- **DevOps, Infrastructure, Tools:** GCP, AWS, Docker, Kubernetes, Terraform, Bazel, Kafka, Ansible, Git
- **IoT, Wearables:** OPC UA, Wear OS Development (Android SDK), Jetpack Compose, Firebase Cloud Messaging (FCM)

EXPERIENCE

- **Daedalus GmbH** Karlsruhe, Germany
Full Stack Engineer 02/2025 to Present

Python, Django, FastAPI, Flask, Gunicorn, React, TypeScript, Tailwind CSS, styled-components, Three.js, React Three Fiber (R3F) and Drei, CadQuery, OpenCascade, Trimesh, Shapely, NumPy, SciPy, Dagster, PostgreSQL, Docker, Kubernetes, GCP, Terraform, Bazel, Kotlin, Wear OS, Jetpack Compose, FCM, OPC UA

Daedalus is an AI-driven precision manufacturing company. Its core product is an entire vertically integrated factory in Karlsruhe - combining FactoryOS, automated workflows, and shopfloor teams to reliably deliver over 12000 high-precision parts monthly.

- **Automated Shopfloor Orchestration and Wearables:** Co-developed an intelligent orchestration system processing about 25000 tasks monthly, effectively automating traditional Shift Leader functions. Designed algorithmic worker-to-task matching based on skills and experience. Architected and delivered a Smart Watches MVP via a custom FCM push-notification microservice to drastically reduce shopfloor response times.
- **Automated Measurements System:** Engineered a core solution for CNC machines to dynamically generate G-code and automatically measure complex part features (lengths, cylinders, holes, spigots). Combined with the Daedalus Proprietary 3D CAM Tool, this system automates about 80% of all feature measurements, saving the equivalent of two full-time shopfloor workers' salaries annually.
- **Daedalus Proprietary 3D CAM Tool:** Architected and built an interactive web-based UI using React and Three.js with GLB file parsing and animated toolpath collision visualization to prevent costly machine damages. Developed from scratch in 6 months, it replicates about 70% of the functionality of expensive enterprise add-ons (such as Siemens NX CAM On-Machine Probing), significantly lowering the technical barrier for shopfloor operators.

- **CAM Workflows and Machine Uptime:** Optimized the process by enabling the automatic embedding of probing cycles directly into NC programs. This innovation accelerated CAM programming speed by 30% and eliminated approximately 1.5 hours of machine downtime per setup.
- **Inspection Data Pipelines:** Streamlined the inspection report auto-finalization logic by migrating it to the backend. This eliminated manual data entry, guaranteed higher measurement precision, and allowed results to be auto-saved directly into the database without worker intervention.

• **DinoLab**

Senior Full Stack Engineer

Tbilisi, Georgia

08/2023 to 01/2025

JavaScript, TypeScript, React, Ruby, Ruby on Rails, Roda, Go, Python, Docker, Kafka, PostgreSQL, Git, HTML, CSS

DinoLab is an educational startup offering an interactive platform for students, parents, and teachers. The platform provides customized online courses in various school subjects. DinoLab helps users develop skills applicable to academics, everyday life, and future careers

- Implemented user action data transmission to Google Analytics, unlocking new business insights into user behavior analysis
- Launched a new mathematics course, increasing revenue by 20% and retention rate by 10%. Led frontend and backend adaptation to microservices architecture and developed the process for translating the course into English
- Configured an additional stage server within a microservices architecture, resulting in reduced Time to Market, enhanced testing quality, and streamlined QA workflows
- Led the development of integration with our EduTech partner, including user registration via SAML protocol and bi-directional data exchange. This resulted in the successful launch of the integration's initial version and comprehensive documentation for ongoing development

• **Uchi.Doma**

Senior Full Stack Engineer

Tbilisi, Georgia

08/2020 to 08/2023

JavaScript, TypeScript, React, Ruby, Ruby on Rails, Docker, Ansible, PostgreSQL, Git, HTML, CSS

Uchi.Doma is an online school that provides high-quality teaching materials, a unique educational platform, and qualified tutors for learning different school subjects

- Refactored the entire codebase to integrate multi-subject functionality on our platform, resulting in a remarkable 300% increase in revenue
- Developed an innovative service for daily database dumping and anonymization in production, reducing database dumping time from 60 to 15 minutes. This initiative not only optimized operations but also elevated our team's data security measures to the next level
- Developed engaging educational games to complement our coding course curriculum, leading to a remarkable 25% increase in student retention rate and a 30% surge in course revenue
- Optimized the Docker image build process for our application by refactoring the Dockerfile, which not only accelerated the build time but also contributed to reducing Time to Market
- Established a new testing server for our QA team, cutting testing time from 30 to 15 minutes. This initiative has empowered the team with greater flexibility and contributed to reducing Time to Market
- Implemented a new design for the client's personal account, enhancing interface responsiveness and user experience. The redesign was met with positive feedback from users and contributed to a more seamless user experience

• **Awatera**

Full Stack Engineer

Moscow

10/2018 to 08/2020

JS, React, OpenUI5, Node.js, Loopback, MongoDB, Git

Awatera is a leading linguistic service and technology provider in the CIS countries. Our internal project, the Translation Management System (TMS), equips our managers with specialized tools to offer competitive pricing for text and voice translations

- Implemented modern React.js as a base UI library on the Frontend, replacing legacy code. This initiative resulted in a smoother user experience, enhanced application performance, and laid a solid foundation for future scalability and development
- Mentored junior developers, facilitating the seamless transition of 3 interns to full-time roles, strengthening the R&D team
- Implemented pair programming techniques within the R&D team, resulting in accelerated feature development, improved code quality, and effective knowledge sharing
- Implemented CI/CD pipelines, significantly reducing code release time and enhancing the efficiency of development cycles
- Reduced server response time by launching second Backend server and fine-tuning nginx balancer

EDUCATION

- **Moscow Institute of Physics and Technology**
Bachelor of Applied Mathematics and Physics

Moscow
09/2013 to 07/2019